Engine Model(s): PT6A

73-10-05

FUEL MANIFOLD AND NOZZLES - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC01-001	Fuel, Engine
PWC03-001	Oil, Engine
PWC03-002	Fluid, Calibrating
PWC05-007	Fluid, Check
PWC05-027	Ink, Marking
PWC05-046	Marker, Ink
PWC05-061	Cloth, Abrasive Coated Crocus
PWC05-145	(See PWC05-019)
PWC09-003	Compound, Sealing
PWC11-013	Compound, Carbon Removing
PWC11-013A	Compound, Carbon Removing
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine
PWC11-049	Remover, Rust
PWC11-061	Rust Remover, Alkaline

3. Special Tools

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30405	Fixture, Leak test	
PWC30416	Puller, Fuel Nozzle Sheath	
PWC30506	Rig, Fuel Manifold Adapters - Flow Test	
PWC30530	Plug, Blanking	
PWC32366	Pusher, Fuel Transfer Tubes	
PWC32811	Fixture, Flow Test	
PWC54246	Puller	Alternative to PWC32366
PWC90086	Spray Chamber	

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

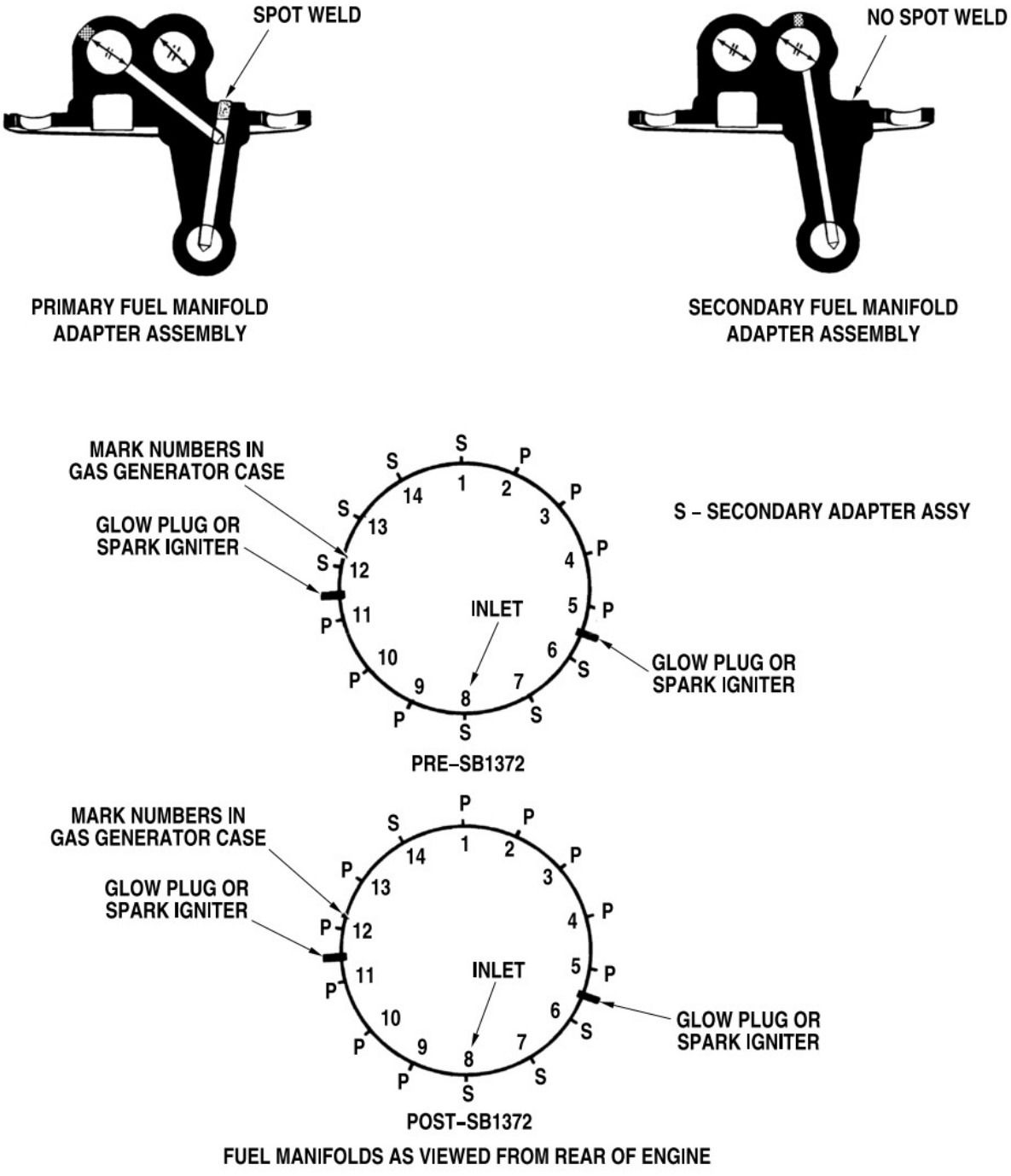
Name	Remarks
Pressure Tester	P/N 2311F
Flushing Fixture	
Stainless Steel Container, Perforated	
Ultrasonic Cleaner	
Blanking Cap	
Blanking Plate	

5. Removal/Installation

A. Procedure Before Removal

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 201 Identification and Location of Fuel Manifold Adapters



c7919c

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

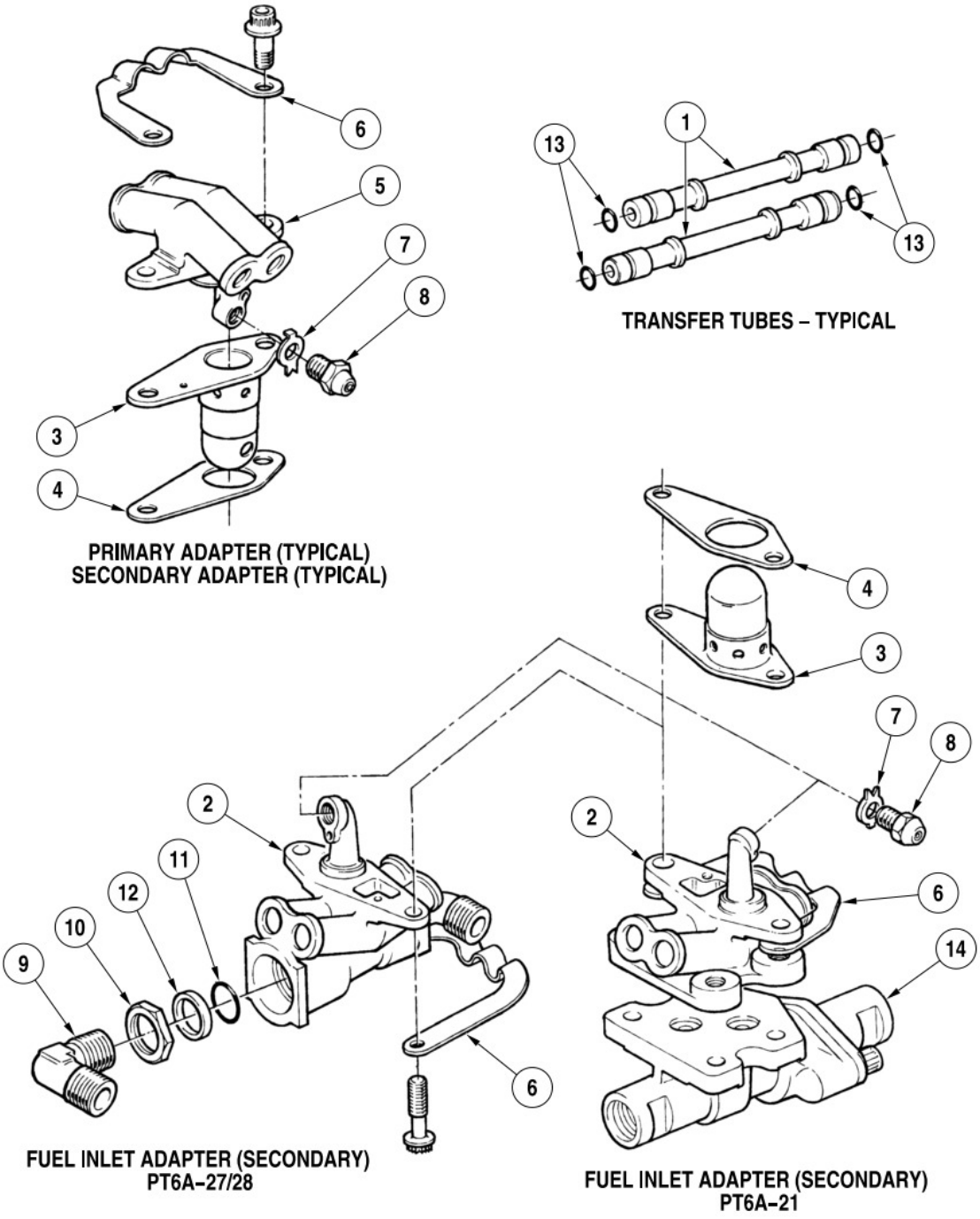
(1) To ease accessibility to the transfer tubes and manifold adapters adjacent to the glow plugs or spark igniters, disconnect the ignition leads (Ref. 74-20-00/74-20-01). Release the ignition lead loop clamps from the support brackets at the center fireseal lower attachment brackets and move the leads clear.Install blanking caps on the glow plugs or spark igniters and lead connectors.

(2) Disconnect both lines from the fuel inlet adapter (PT6A-27/-28) or flow divider (PT6A-21) and install blanking caps.

B. Removal of Fuel Manifold Adapters

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 202 Removal/Installation of Fuel Manifold Adapters and Nozzle Assemblies



c7918a

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Fuel Transfer Tube
2. Fuel Manifold Inlet Adapter
3. Sheath
4. Gasket
5. Fuel Manifold Adapter
6. Lockplate
7. Keywasher
8. Fuel Nozzle
9. Elbow
10. Locknut
11. Preformed Packing
12. Back-up ring
13. Preformed Packing
14. Flow Divider and Dump or Purge Valve

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: IF THE ENGINE IS NOT SEPARATED AT FLANGE C AND IT IS NECESSARY TO REMOVE ALL THE FUEL MANIFOLD ADAPTERS, THEN MAKE SURE THAT THE GLOW PLUGS OR SPARK IGNITERS STAY INSTALLED. IF NOT, THERE COULD BE ALIGNMENT PROBLEMS WITH THE COMBUSTION CHAMBER OUTER LINER WHEN YOU INSTALL THE FUEL MANIFOLD ADAPTERS AGAIN.

NOTE: The following procedure itemizes a removal sequence commencing with the No. 8 fuel manifold inlet adapter, and which is best achieved by consideration of the Nos. 7, 8 and 9 adapters as a group. The procedural sequence may be modified by the operator as convenient for adapters at other locations.

(1) Using a suitable dye marker (PWC05-027) or (PWC05-046), number the position of each manifold adapter to identify its original location for reinstallation (Ref. Fig. 201) and to aid detecting hot section damage.

(2) Remove bolts securing transfer tube locking plate (6, Fig. 202) and inlet manifold adapter (2) to gas generator case. Remove locking plate (6).

(3) Remove bolts securing locking plates (6) to the primary and secondary manifold adapters (5) adjacent to inlet manifold adapter (2). Remove locking plates (6).

WARNING: PUT ON CLEAN PROTECTIVE GLOVES WHEN YOU USE THE PUSHER (PWC32366) TO PREVENT INJURY TO YOUR FINGERS.

CAUTION: MOVE THE TRANSFER TUBES AWAY TO PREVENT THE DAMAGE TO THE STRAINER ELEMENTS WHICH YOU CAN FIND IN THE BORE OF THE DIAMETER.

(4) Hold all the three adapters, then move the interconnecting fuel transfer tubes (1) into adapter bores (5). Use the puller (PWC54246) or pusher (PWC32366) to move the fuel transfer tubes in a clockwise direction, away from the inlet manifold adapter bores.

WARNING: PUT ON CLEAN PROTECTIVE GLOVES WHEN YOU USE THE PUSHER (PWC32366) TO PREVENT INJURY TO YOUR FINGERS.

(5) Remove the inlet manifold adapter (2) (with flow divider and dump valve (14) installed). Use the pusher (PWC32366) or puller (PWC54246) to remove the fuel transfer tubes (1) from the adjacent adapters (5). Remove and discard preformed packings (13) from the transfer tubes.

NOTE: Do not remove flow divider and dump valve from inlet manifold adapter unless cleaning is necessary (Ref. Cleaning/Painting). You can do the below given inspection and testing procedures with the parts can stay attached as an assembly.

(6) Remove the gasket (4) from the sheath (3) on the inlet manifold adapter (2).

(7) Remove the remaining manifold adapters (5) progressively from the gas generator case by removing bolts and locking plates (6). As each adapter is removed, withdraw interconnecting fuel transfer tubes (1). Remove and discard preformed packings (13). Remove metal gaskets (4) from sheaths (3) on the adapters.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: DO NOT PRY THE SHEATHS OFF WITH A SCREWDRIVER.

(8) Remove the sheaths (3) from the manifold adapters using puller (PWC30416) if sheath is tight fitting on adapter boss.

CAUTION: EXTREME CARE MUST BE EXERCISED WHEN HANDLING THE FUEL NOZZLE ASSEMBLIES SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT-FREE COTTON GLOVES OR SURGICAL GLOVES SHOULD BE WORN AT ALL TIMES WHEN HANDLING THESE PARTS.

CAUTION: MAKE SURE WRENCH SOCKET IS CORRECTLY ENGAGED ON NOZZLE DURING INSTALLATION. FUEL NOZZLE OUTLET MAY BE DAMAGED IF WRENCH SLIPS.

(9) Straighten the lugs on the keywashers (7) and remove the nozzle assemblies (8) from the adapters (2) and (5). Discard the keywashers.

(10) Note the relative angle of each elbow to the inlet manifold adapter (2) (starting control installation only) and remove the elbows (9) and locknuts (10). Discard preformed packings (11) and back-up rings (12).

(11) To prevent exposure to dust and dirt, place the manifold components in clean, covered containers or polyethylene bags until required for use.

(12) Send rejected assemblies for repair or overhaul in original packaging to prevent parts contacting each other during shipment.

C. Installation of Fuel Manifold Adapters

CAUTION: USE OF ENGINE OIL (PWC03-001) IS ONLY PERMITTED AS A LUBRICANT FOR INSTALLATION OF THE FUEL NOZZLE ASSEMBLY. DO NOT USE CALIBRATION FLUID OR OTHER LUBRICANTS.

CAUTION: EXTREME CARE MUST BE EXERCISED WHEN HANDLING THE FUEL NOZZLE ASSEMBLIES SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT-FREE COTTON GLOVES OR SURGICAL GLOVES SHOULD BE WORN AT ALL TIMES WHEN HANDLING THESE PARTS.

CAUTION: MAKE SURE TO CORRECTLY ENGAGE THE WRENCH SOCKET ON THE NOZZLE DURING INSTALLATION. IF WRENCH SLIPS, FUEL NOZZLE OUTLET CAN DAMAGE. HOLD THE STEM WHEN YOU INSTALL THE NOZZLE TO PREVENT BENDING.

CAUTION: USE OF THE ENGINE OIL (PWC03-001) IS ONLY PERMITTED AS A LUBRICANT FOR INSTALLATION OF THE FUEL NOZZLE TIP ASSEMBLIES. DO NOT USE CALIBRATION FLUID OR OTHER LUBRICANTS.

CAUTION: DO NOT TORQUE MORE THAN THE LIMIT TO ALIGN FLAT ON TIP WITH KEYWASHERS.

(1) With a 10X magnifying glass verify that each manifold adapter assembly carries the correct

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

detail fuel nozzle assembly tip part number (Ref. Illustrated Parts Catalog).

(2) Install the elbows (9) on the inlet adapter (starting control installation only) (Ref. 70-00-00, REMOVAL/INSTALLATION) in the same positions as noted on removal (Ref. Subpara. B.(10)).

(3) Install the fuel nozzle (8, Fig. 202) in the fuel manifold adapters (2) and (5) with the new keywashers (7) at each location. Use engine oil (PWC03-001) and torque the nozzle assemblies 45 to 50 lbf. in. Do not lock the keywashers until the test complete.

(4) Leak test each nozzle and adapter assembly (Ref. Adjustment/Test). Repair or replace nozzles that leak (Ref. Repair).

(5) Function test each nozzle and adapter assembly (Ref. Adjustment/Test).

CAUTION: DO NOT USE SHARP EDGED TOOLS TO BEND OR SET KEYWASHER TABS. TAB DETACHMENT MAY OCCUR WITH SUBSEQUENT DAMAGE TO THE NOZZLE TIP.

(6) On completion of satisfactory tests, lock each keywasher on respective nozzle assembly. Do not exceed specified torque to align flat on tip with keywasher.

(7) Do a leak test check of each fuel nozzle and adapter assembly (Ref. Adjustment/Test). If leaks are found, replace the keywasher (7) and do Steps (3) thru (5) again.

(8) Install the manifold adapters and fuel transfer tubes as follows:

NOTE: Primary fuel manifold adapters are identified by a single weld blob on the larger mounting flange. Other weld blobs appearing on the knuckle section of the adapters should be ignored

(a) Assemble sheaths (3) on all adapters (2 and 5). Make sure each locating pin engages hole in each sheath.

(b) With nozzle adapter and sheath pressed together by hand, check clearance between adapter and sheath flanges. Maximum gap allowed is 0.003 inch (Ref. Fig. 203). Larger gap suggests either or both parts are distorted. Send parts to P&WC (Ref. INTRODUCTION, Customer Support) for repair.

(c) Carefully check gap between fuel nozzle tip and side of hole in sheath; clearance of 0.020 inch is required (Ref. Fig. 203). If clearance is less at any point, either or both parts are distorted. Send parts to P&WC (Ref. INTRODUCTION, Customer Support) for repair.

(d) Lubricate and install preformed packings (13, Fig. 202) on all fuel transfer tubes (1) using a thin layer of engine oil (PWC03-001). Position fuel transfer tubes into ports on one side of manifold adapters (2 and 5). Fully insert fuel transfer tubes in their respective ports until the bottom of manifold adapter is reached.

CAUTION: TO AVOID POSSIBLE FAILURE OF THE STAINLESS STEEL GASKETS, IT IS ADVISABLE TO SLIDE THE GASKET OVER THE SHEATH, CAREFULLY ALIGN THE HOLES AND INSERT BOTH BOLTS. THE WHOLE ASSEMBLY CAN THEN BE MATED WITH THE PADS ON THE GAS GENERATOR CASE, THUS AVOIDING ANY ATTEMPT TO LEVER THE GASKETS INTO ALIGNMENT WITH THE BOLTS.

(e) Position the Pre-SB1276/Post-SB1276 gasket (4) over the sheath (3) on the inlet manifold

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

adapter (2) and align the bolt holes. The sheath flange must be flat.

NOTE: 1. The gasket may be put on either side. For consistency, all gaskets should be installed with the flat side against the gas generator.

NOTE: 2. Post-SB1167: For engines with the conversion coated gas generator case only, lightly coat both faces of the gasket (4) with corrosion-preventive compound (PWC09-003).

(f) Position the gaskets (4) over the sheaths (3) on the primary and secondary manifold adapters (5) and align on the bolt holes.

WARNING: PUT ON CLEAN PROTECTIVE GLOVES WHEN YOU USE THE PUSHER (PWC32366) TO PREVENT INJURY TO YOUR FINGERS.

(g) Start with each side of the inlet manifold adapter (2). Install the remaining manifold adapters (5) and sheaths (3) on the gas generator case at the locations specified before. Use the pusher (PWC32366) or puller (PWC54246) to engage fuel transfer tubes (1) to interconnect with the adjacent adapter at each location.

NOTE: To find the hot section damage easily, install the replacement nozzle at the initial locations from which unserviceable item was removed. If the cleaning of the nozzles after the flow test is not necessary, then install the nozzles at the initial locations (Ref. Fig. 201).

(h) Install locking plate (6) and bolts to the gas generator case . Torque bolts fingertight.

(9) When all remaining manifold adapters (5) are positioned, remove bolts from inlet manifold adapter (2). Assemble the locking plate (6) and reinstall the bolts.

CAUTION: MAKE SURE ALL 14 MANIFOLD ADAPTER LOCKING PLATES ARE CORRECTLY INSTALLED.

CAUTION: PRIOR TO TIGHTENING BOLTS OR LOCKNUTS, CHECK SEATING OF METAL GASKET RELATIVE TO SHEATH AND BOSS. TIGHTENING SHOULD BE DONE EVENLY ON EACH ADAPTER TO PROVIDE FULL SEATING POTENTIAL OF METAL GASKET.

(10) Tighten all adapter mounting bolts, in a sequence, 15 to 20 lb.in. Retighten 32 to 36 lb. in. in the same sequence. Secure bolts with lockwire.

NOTE: After torquing, a 0.001 in. gap is allowed between the adapter and sheath flanges (Ref. Fig. 203).

(11) Remove the blanking caps from the fuel delivery lines and connect the lines to the elbows. Tighten the coupling nuts 90 to 100 lb.in. and fasten with lockwire.

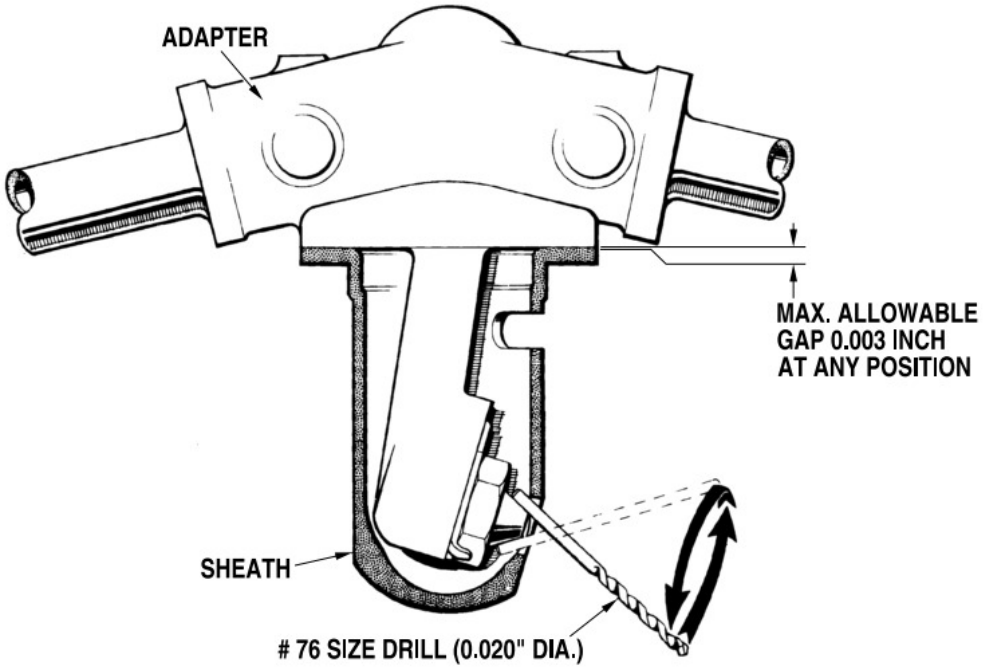
(12) Remove the blanking caps from the spark igniters and from harness leads and connect the leads to the igniters. Tighten the connections fingertight, plus 45 degrees and fasten with lockwire. Secure the ignition lead loop clamps to the support brackets at the center fireseal lower mounts and tighten the nuts 32 to 36 lb.in.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

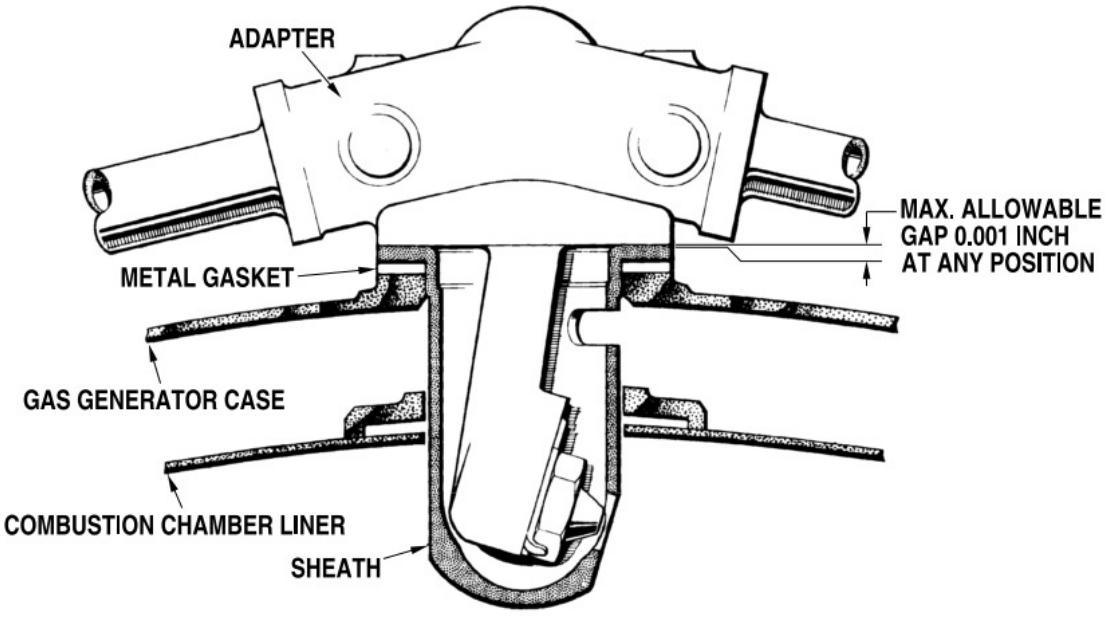
(13) Check function of the fuel manifold installation (Ref. 71-00-00, POWER PLANT - ADJUSTMENT/TEST).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 203 Fuel Manifold Installation - Clearance Checks



SECTION THROUGH FUEL MANIFOLD ADAPTER AND SHEATH HAND HELD



SECTION THROUGH FUEL MANIFOLD ADAPTER AND SHEATH
INSTALLED ON GAS GENERATOR CASE

c41914

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

6. Cleaning/Painting

WARNING: CARBON REMOVING COMPOUND IS TOXIC AND CAUSTIC, AND MUST BE HANDLED WITH EXTREME CARE. AVOID ALL DIRECT CONTACT WITH SKIN OR CLOTHING. PREVENT CONTACT WITH EYES AND AVOID INHALATION OF THE VAPORS. COMPOUNDS CAN BE FATAL IF SWALLOWED. IF CONTACT WITH EYES OR SKIN IS MADE, WASH AFFECTED AREAS THOROUGHLY FOR 15 MINUTES WITH WATER AND RINSE WITH SATURATED BORIC ACID SOLUTION. IF ACCIDENTALLY SWALLOWED, GET MEDICAL ATTENTION IMMEDIATELY. CONSULT WITH PHYSICIAN AS RAPIDLY AS POSSIBLE FOR ALL CONTACT CASES.

CAUTION: ONLY USE RECOMMENDED ALKALINE CLEANING SOLUTION WITH A PH GREATER THAN 7. SOLUTIONS WITH A PH VALUE LOWER THAN 7 (ACIDIC) MAY CAUSE FUEL NOZZLE DAMAGE.

CAUTION: OBSERVE ALL FIRE AND SAFETY PRECAUTIONS WHEN FUELS OR SIMILAR COMBUSTIBLES ARE USED.

A. Cleaning of Fuel Manifold Adapter Assemblies

(1) Remove fuel flow divider from inlet adapter (Ref. 73-10-04, Removal/Installation).

(2) Remove fuel manifold sheaths from adapter assemblies.

CAUTION: MAKE SURE ADAPTERS ARE SEPARATED FROM EACH OTHER TO PREVENT DAMAGE DURING CLEANING.

(3) Place as many manifold adapter assemblies as possible in a perforated stainless steel container constructed to prevent contact between parts.

NOTE: Wires may be used to suspend parts in cleaning solution to prevent contact during cleaning.

(4) Prepare caustic solution by diluting 2 to 3 pounds of carbon removing compound (PWC11-013) or (PWC11-049) in one (1) U.S. gallon of water (3.8 liters) and heat to approximately 90°C (200°F). Pour caustic solution/cleaner into a tank of ultrasonic cleaner.

NOTE: 1. 2 to 3 lb. carbon removing compound (PWC11-013) or (PWC11-049) per (US) gallon produces 24 to 36 % solution.

NOTE: 2. Carbon removing compound (PWC11-013A) or alkaline rust remover (PWC11-061) can be used as an alternative to carbon removing compound (PWC11-013).

(5) Suspend the perforated stainless steel container containing manifold adapter assemblies in the cleaner tank solution. Make sure that all manifold adapters are completely immersed.

(6) Cover tank and activate ultrasonic cleaner for approximately one hour.

WARNING: PUT ON PROTECTIVE GLOVES WHEN HANDLING HOT PARTS TO PREVENT PERSONAL INJURY.

(7) After you clean the assemblies, immediately flush the assemblies fully in very hot water at

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

82°C (180°F) for two to three minutes.

WARNING: PUT ON PROTECTIVE GLOVES WHEN HANDLING HOT PARTS TO PREVENT PERSONAL INJURY.

CAUTION: FLUSH THE ASSEMBLIES IMMEDIATELY AFTER THE CLEANING TO PREVENT DAMAGE TO THE ASSEMBLIES BY THE REMAINING CARBON REMOVER SOLUTION.

(8) Flush the parts again in very hot water at 82°C (180°F) for two to three minutes.

(9) For PT6A-21 Engines:

(a) Install blanking plate (Ref. Fig. 204) and preformed packings on mounting face of fuel inlet manifold adapter.

(b) Install blanking tube assemblies (18, Fig. 211) at three locations on fuel inlet manifold adapter.

(c) Connect tube assembly (9, Fig. 211) at fourth location and flow fixture (PWC32811) to fuel inlet.

(d) Flow clean with hot water at normal tap pressure, 30 psig max through fuel inlet manifold adapter for a minimum of one minute.

(e) Remove blanking plate (Ref. Fig. 204). tube assembly, three blanking tube assemblies (18, Fig. 211) and preformed packings. Discard preformed packings.

(10) For PT6A-27/28 Engines:

(a) Install applicable blanking caps at five locations on fuel inlet manifold adapter (Ref. Fig. 202).

(b) Connect tube assembly (9, Fig. 211) at sixth location and flow fixture (PWC32811) to fuel inlet.

(c) Flow clean with hot water at normal tap pressure, 30 psig max through fuel inlet manifold adapter for a minimum of one minute.

(d) Remove blanking caps (Ref. Fig. 202) from five locations and tube assembly.

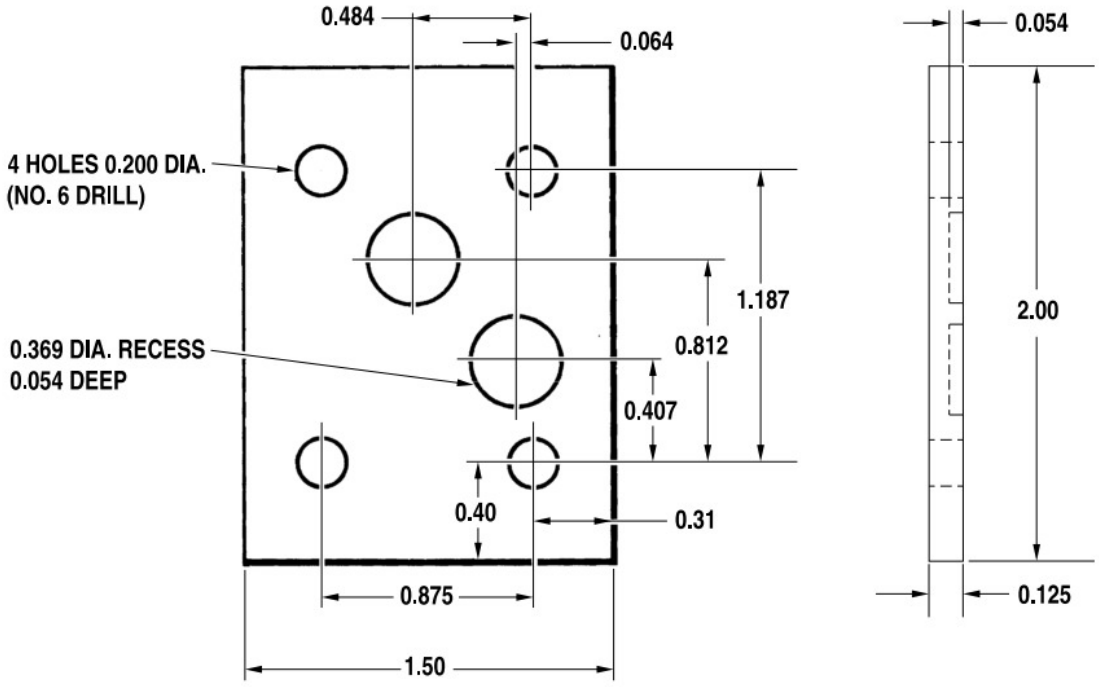
(11) Immediately dry assemblies thoroughly using clean, filtered compressed air or nitrogen.

(12) Do leak and functional test (Ref. Para. 9.B. and C.).

NOTE: Before commencing the tests as described, allow fuel to flow through the nozzle and manifold adapter assembly for at least one minute at 30 psig.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 204 Inlet Adapter Blanking Plate



MATERIAL: MILD STEEL OR ALUMINUM SHEET 0.125 THICK (MIN.)

LOCAL MANUFACTURE

c3790c

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

B. Cleaning of Fuel Nozzles (Removed from Fuel Manifold Adapters)

CAUTION: EXTREME CARE MUST BE USED WHEN THE FUEL NOZZLE ASSEMBLIES ARE HANDLED SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT-FREE COTTON GLOVES OR SURGICAL GLOVES SHOULD BE WORN AT ALL TIMES WHEN THESE PARTS ARE HANDLED.

(1) Place fuel nozzles in perforated steel container, and clean using same procedures as that for fuel manifold adapter assemblies (Ref. Subpara. A.).

CAUTION: FLUSHING MUST BE DONE IMMEDIATELY AFTER THE CLEANING PROCESS TO AVOID DAMAGE TO THE NOZZLES BY RESIDUAL CARBON REMOVER SOLUTION.

(2) Install nozzles in flushing fixture (Ref. Fig. 205).

NOTE: Use suitable blanking plugs if all 14 positions are not filled.

(3) Connect the hot water supply to the flushing fixture and flow clean, very hot water through the nozzles for one minute at normal tap water pressure up to 30 psig.

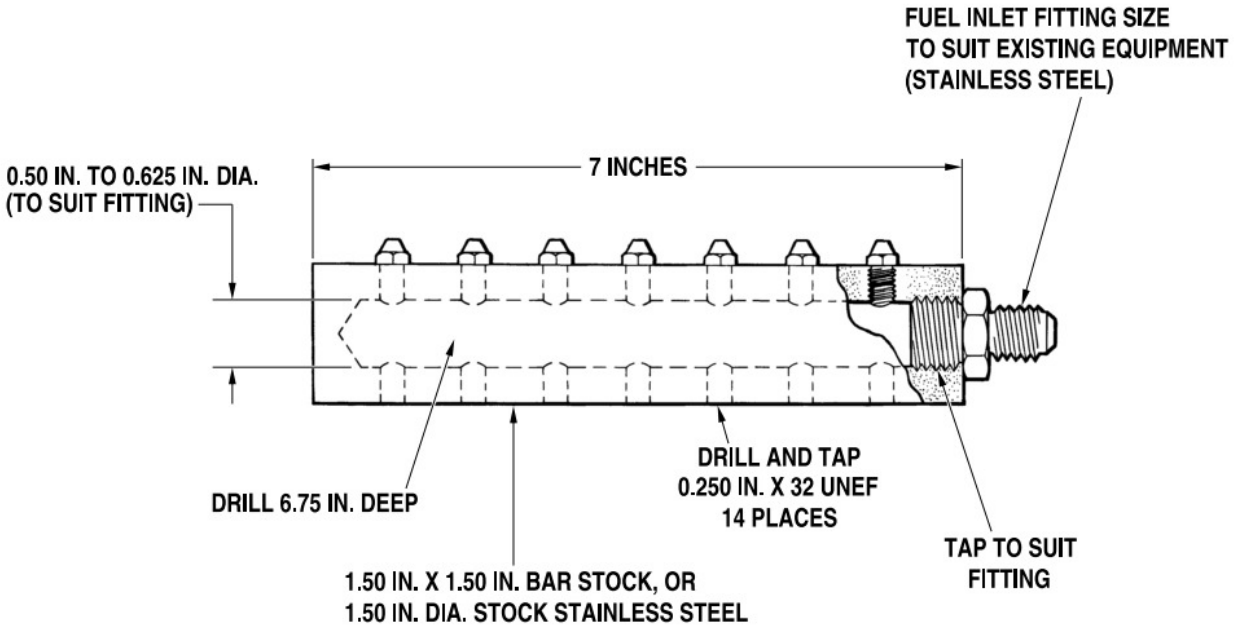
(4) Disconnect the flushing fixture from the hot water supply, connect to a supply of clean filtered compressed air or nitrogen and dry thoroughly. Disconnect the flushing fixture from the air supply.

(5) Connect the flushing fixture to the fuel manifold adapter assembly test rig and flow fuel through the nozzles for one minute at 30 psig.

(6) Remove the nozzles from the flushing fixture and store in a clean container until required for reinstallation and test in the adapter assemblies.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 205 Fuel Nozzle Flushing Fixture



LOCAL MANUFACTURE

c3812a

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

C. Cleaning Fuel Manifold Adapter (Nozzles Removed) and Adapter Sheaths

CAUTION: EXTREME CARE MUST BE USED WHEN THE FUEL NOZZLE ASSEMBLIES ARE HANDLED SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT-FREE COTTON GLOVES OR SURGICAL GLOVES SHOULD BE WORN AT ALL TIMES WHEN THESE PARTS ARE HANDLED.

NOTE: SPOP 218 can be used as an optional cleaning procedure for fuel manifold adapter sheath only.

(1) Place the fuel manifold adapters and sheaths in the perforated stainless steel container, and proceed as described in Subpara. A., steps (4) through (7). Reduce cleaning time to 30 minutes.

CAUTION: FLUSH WASHING MUST BE DONE IMMEDIATELY AFTER THE CLEANING PROCESS.

(2) Pressure wash parts thoroughly to remove residual compound under very hot, running water for at least one minute. Make sure that adequate water flow is maintained through the internal bores of the adapters. Remove any remaining loose carbon build-up on the sheaths by scrubbing with a non-metallic bristle brush.

(3) Immediately dry parts thoroughly using clean, filter compressed air.

(4) Install the fuel nozzles in the manifold adapters (Ref. Removal/Installation) and test (Ref. Adjustment/Test).

7. Inspection/Check

CAUTION: USE EXTREME CARE WHEN HANDLING FUEL NOZZLE ASSEMBLIES SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT FREE GLOVES SHOULD BE WORN AT ALL TIMES WHEN HANDLING PARTS.

A. Fuel Nozzle Assembly

(1) Check the nozzle assembly for burrs and similar defects. Threads and hexagon may be lightly stoned to remove nicks and burrs.

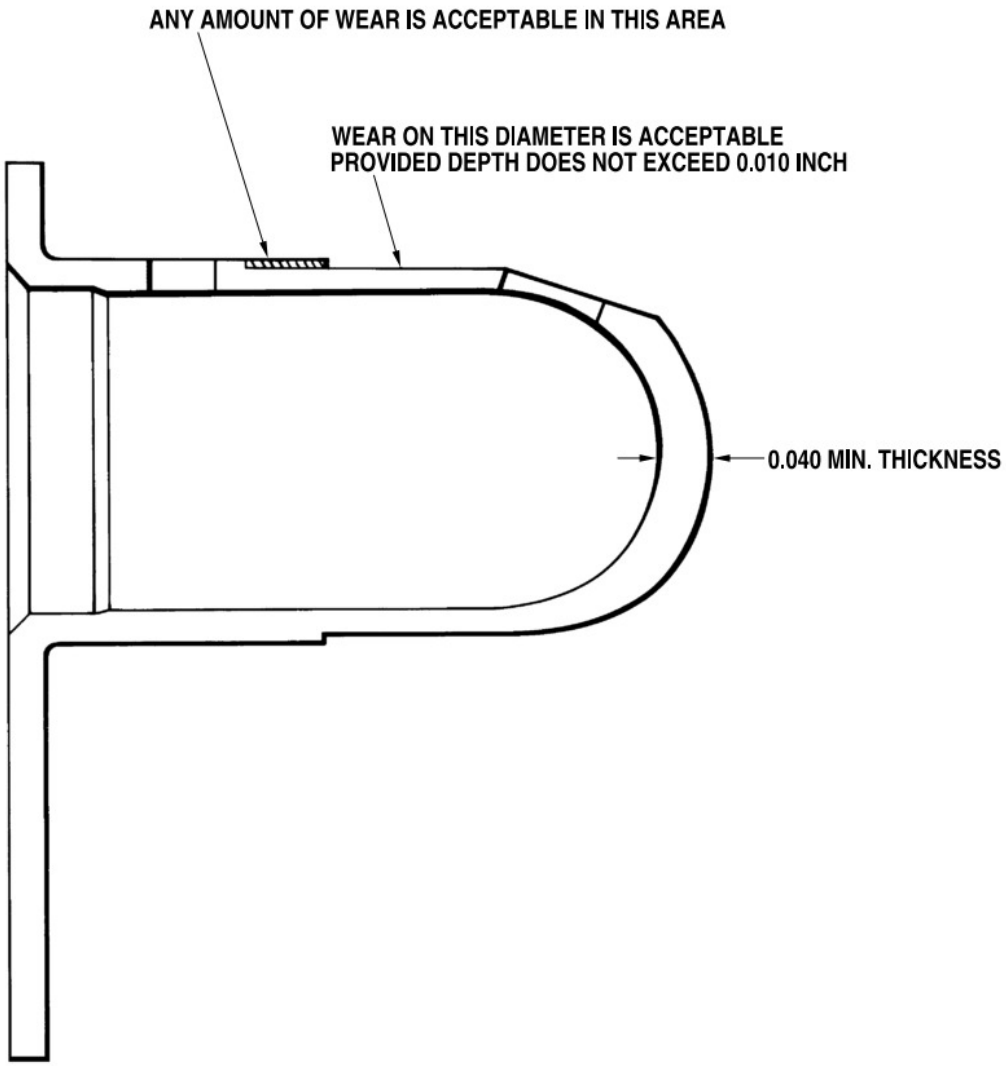
B. Nozzle Sheath (Ref. Fig. 206)

(1) Erosion, any loss of coating is acceptable provided any loose coating is removed by light buffing and 0.040 inch minimum dome thickness is retained.

(2) Fretting wear on outer diameter is permitted, if it is not more than 0.010 inch (Ref. Fig. 206).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 206 Nozzle Sheath Wear Limits



c8403

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

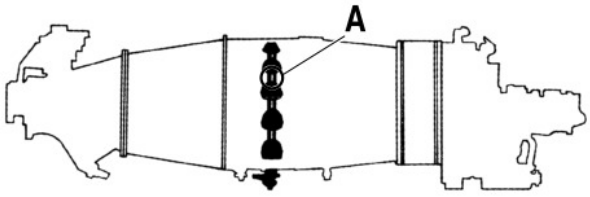
C. Manifold Adapter

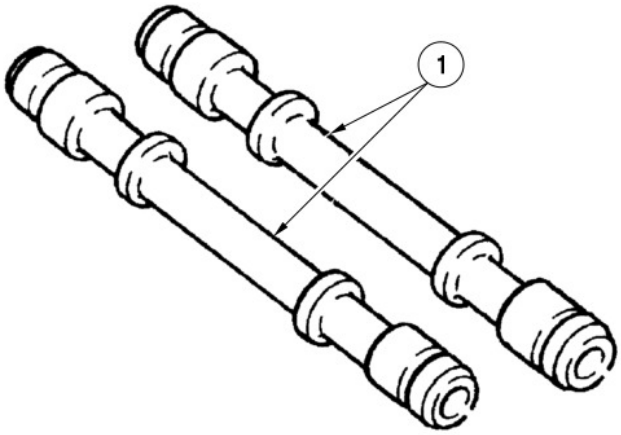
(1) Check the locating pin for security in the adapter and for burrs and similar defects. Check threads for damage.

D. Transfer Tubes (Ref. Fig. 207)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 207 Fuel Manifold Transfer Tube





DETAIL A
(TYPICAL)

c210740

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Transfer Tube

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(1) Examine for general condition and check preformed packing grooves and lands for damage. Minor surface damage in the form of burrs and scratches, may be removed by blending.

(2) Inspect for bent transfer tubes (1) as follows:

(a) Put the transfer tube on flat surface and roll the transfer tube. If an elliptical motion is produced at any point along the tube, reject the tube.

(3) Inspect condition of external anodic finish.

(4) Examine the bore of each tube is free from blockage. Install the cap on the transfer tube (1) after inspection.

8. Approved Repairs

CAUTION: REPAIRS ARE NOT PERMITTED ON NOZZLE ORIFICE FACE.

A. Procedure

CAUTION: DAMAGE MAY OCCUR TO THE ADAPTER SEALING FACE IF USING TOO COARSE A GRIT OF ABRASIVE CLOTH OR APPLICATION OF UNEQUAL PRESSURE ON ADAPTER FACE DURING LAPPING. THIS PROCEDURE IS ONLY RECOMMENDED FOR ADAPTERS THAT ARE FOUND TO BE LEAKING AFTER INSTALLATION OF A NOZZLE TIP.

CAUTION: IF LEAKAGE CANNOT BE STOPPED BY LAPPING OF THE ADAPTER SEALING FACE, DO NOT OVERTORQUE NOZZLE TIP TO ACHIEVE CORRECT SEALING. RETURN DEFECTIVE NOZZLE ASSEMBLY TO AN AUTHORIZED OVERHAUL SHOP FOR REPAIR.

(1) Repair of adapters, sheaths and nozzle assemblies is limited to local blending of minor defects such as scores, nicks, scratches and gouges on exterior surfaces.

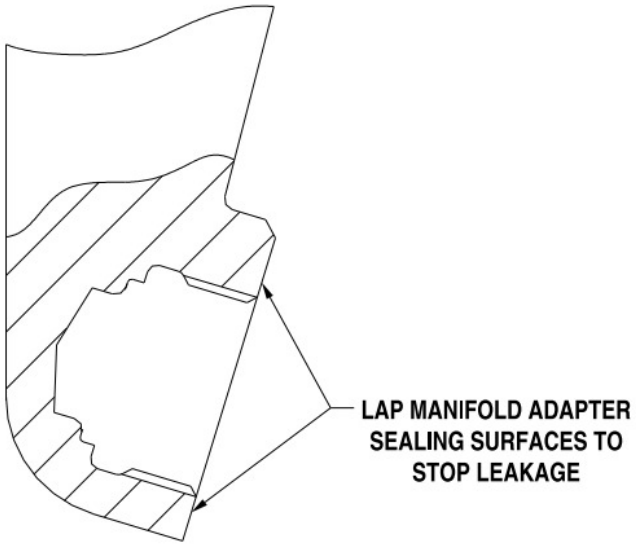
(2) Clean up minor defects by blending with a fine stone or file and/or by polishing with crocus cloth (PWC05-061). Thread damage on adapters may be repaired with a suitable Swiss file, thread chaser or appropriate die.

(3) Remove all sharp edges and high spots. Lap adapter sealing surfaces (Ref. Fig. 208) against a flat base (± 0.0002 inch flatness). Use compound (PWC05-145) or an abrasive cloth (PWC05-061) lubricated with a drop of fuel.

(4) Clean all parts after repair by pressure washing in petroleum solvent (PWC11-027) or (PWC11-031).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 208 Manifold Adapter Sealing Surfaces



c67072

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

9. Adjustment/Test

CAUTION: OBSERVE FIRE SAFETY PRECAUTIONS AT ALL TIMES WHEN PROCEDURES INVOLVE THE USE OF FUELS OR SIMILAR COMBUSTIBLES.

CAUTION: DAMAGE TO THE HOT SECTION MAY BE CAUSED BY DEFECTIVE FUEL NOZZLES. REGULAR NOZZLE INSPECTION AND CLEANING IS RECOMMENDED TO EXTEND HOT SECTION LIFE.

A. General

(1) Definitions of terms used in the text to describe specified test conditions for nozzles (Ref. Fig. 209):

(a) "Streakiness" is defined as variation in spray quantity between different parts of spray cone and appearing as lighter or darker streaks in spray.

1 A total of 20% of fuel spray may show light streaks.

2 Caused by carbon deposit at nozzle face.

3 Brush nozzle surface during flow test.

(b) "Spitting" is a condition which exists when large drops of unatomized fuel occur intermittently and usually on outside of spray cone.

1 No spitting allowed.

2 Caused by carbon deposit at nozzle face.

3 Brush nozzle during flow test.

(c) "Drooling" is a condition which occurs when large drops of unatomized fuel form on nozzle face.

1 No drooling allowed.

2 Caused by carbon deposit at nozzle orifice or by partial obstruction of fuel nozzle distributor.

3 Brush nozzle surface during flow test and ultrasonically clean.

(d) "Void" area of fuel spray showing discontinuity in fuel flow (air gap).

1 No void allowed.

2 Caused by obstruction on internal fuel passage.

3 Ultrasonically clean.

(e) "Skewness" describes a spray condition that is not centered.

1 No skewness allowed.

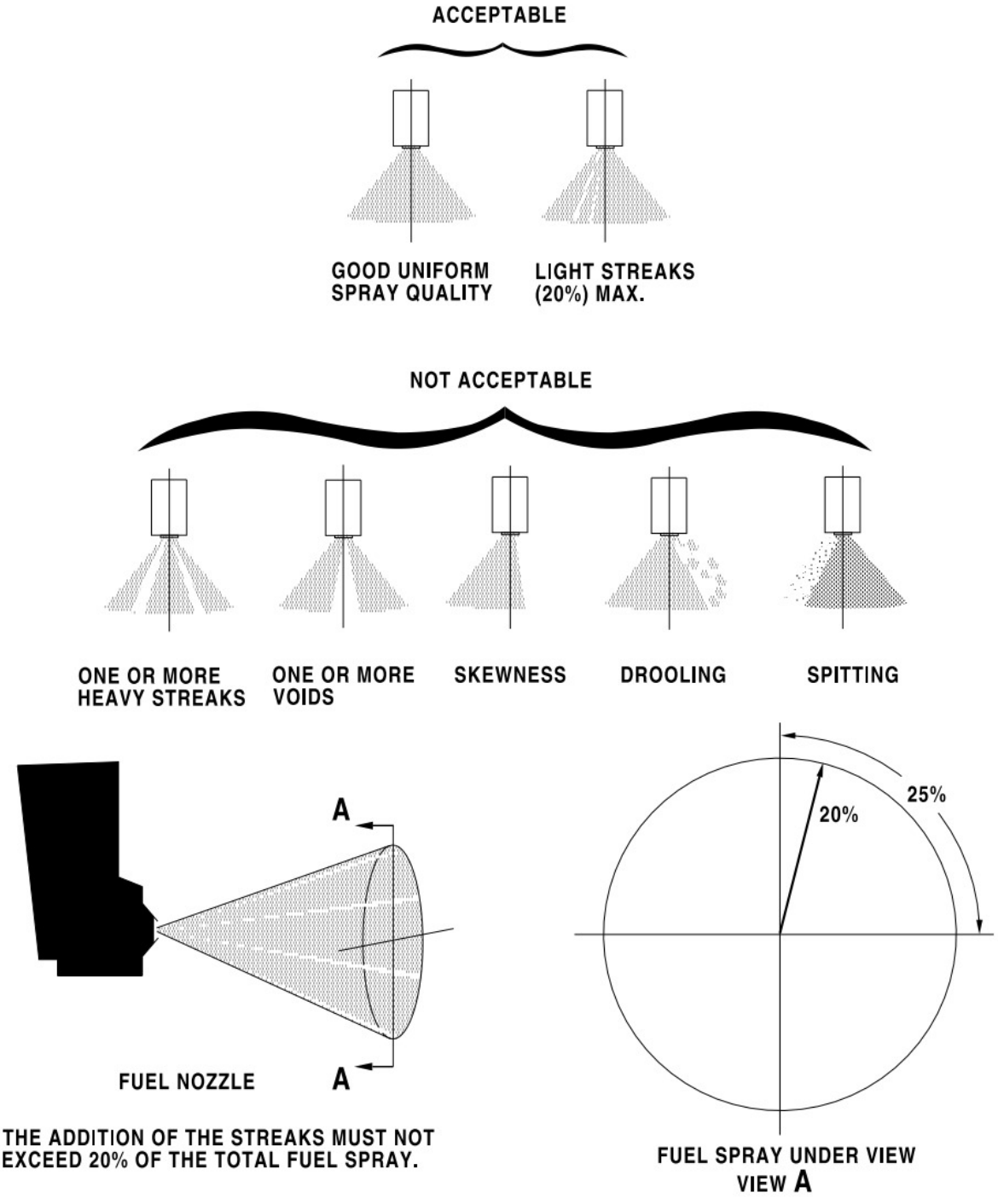
2 Caused by damage to nozzle orifice.

3 Not repairable at field level.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 209 Fuel Nozzle Spray Conditions

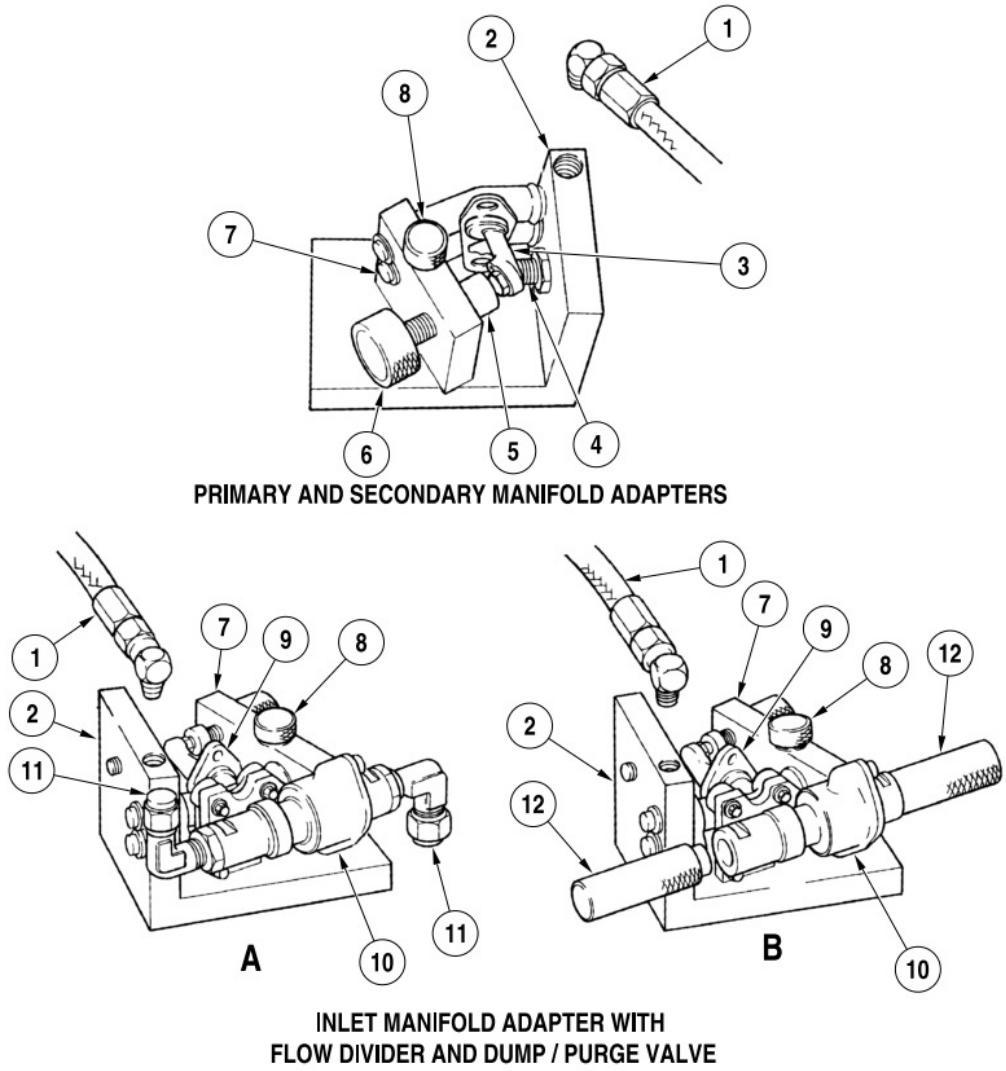


c67153a

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

B. Leakage Test of Fuel Manifold Adapters (Ref. Fig. 210)

Figure 210 Manifold Adapter and Nozzle Assembly Leakage Test



c1881

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Hose Assembly
2. Test Fixture (PWC30405)
3. Primary or Secondary Manifold Adapter and Nozzle Assembly
4. Setscrew (Part of Fixture)
5. Plastic Pad (Part of Fixture)
6. Torque Screw (Part of Fixture)
7. Pivot Block (Part of Fixture)
8. Pivot Screw (Part of Fixture)
9. Inlet Manifold Adapter and Nozzle Assembly
10. Flow Divider and Dump Valve
11. Blanking Cap
12. Plug

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: USE EXTREME CARE WHEN HANDLING FUEL NOZZLE ASSEMBLIES SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT FREE COTTON GLOVES OR SURGICAL GLOVES SHOULD BE WORN AT ALL TIMES WHEN HANDLING PARTS.

(1) Loosen screws (8, 6 and 4) of test fixture (2) and remove pivot block (7).

(2) Insert plugs of pivot block (7) into ports in the nozzle assembly. Make sure the preformed packings are not pinched during nozzle installation.

(3) With plugs fully inserted into adapter ports, hold parts firmly and tighten pivot screw (8).

(4) Blank off nozzle orifice:

(a) Turn setscrew (4) until it just makes contact with rear of adapter behind nozzle.

(b) Turn torque screw (6) until plastic pad (5) seats on nozzle face.

(c) Tighten setscrew (4) and screw (6), simultaneously, to make sure that plastic pad (5) closes nozzle orifice without distortion of adapter.

(d) Tighten locknut on setscrew (4).

(5) When inlet manifold adapter and nozzle assembly (9) with attached flow divider (10) are to be tested, do the following additional steps:

(a) Blank off elbow on flow divider (10) with caps (11). Tighten caps 40 to 45 lb.in. (Ref. View A).

(b) If elbows are not installed, blank off ports in flow divider (10) with plugs (12, PWC30530) (Ref. View B).

(6) Check for external leaks between nozzle and adapter using one of the following methods:

(a) Method A: Connect hose assembly (1) to a supply of clean, dry compressed air or nitrogen and apply 200 psig to test fixture (2). Check for leaks using leak check fluid (PWC05-007) or by immersing in solvent (PWC11-027) or (PWC11-031). No leaks are permitted.

(b) Method B: Fill the hose assembly (1) with the fuel (PWC01-001) or calibrating fluid (PWC03-002). Apply 500 psig of compressed air or nitrogen to the hose assembly (1) to pressurize the fuel. Hold the hose assembly (1) vertical and examine the fuel nozzle for leakage. No leaks are permitted.

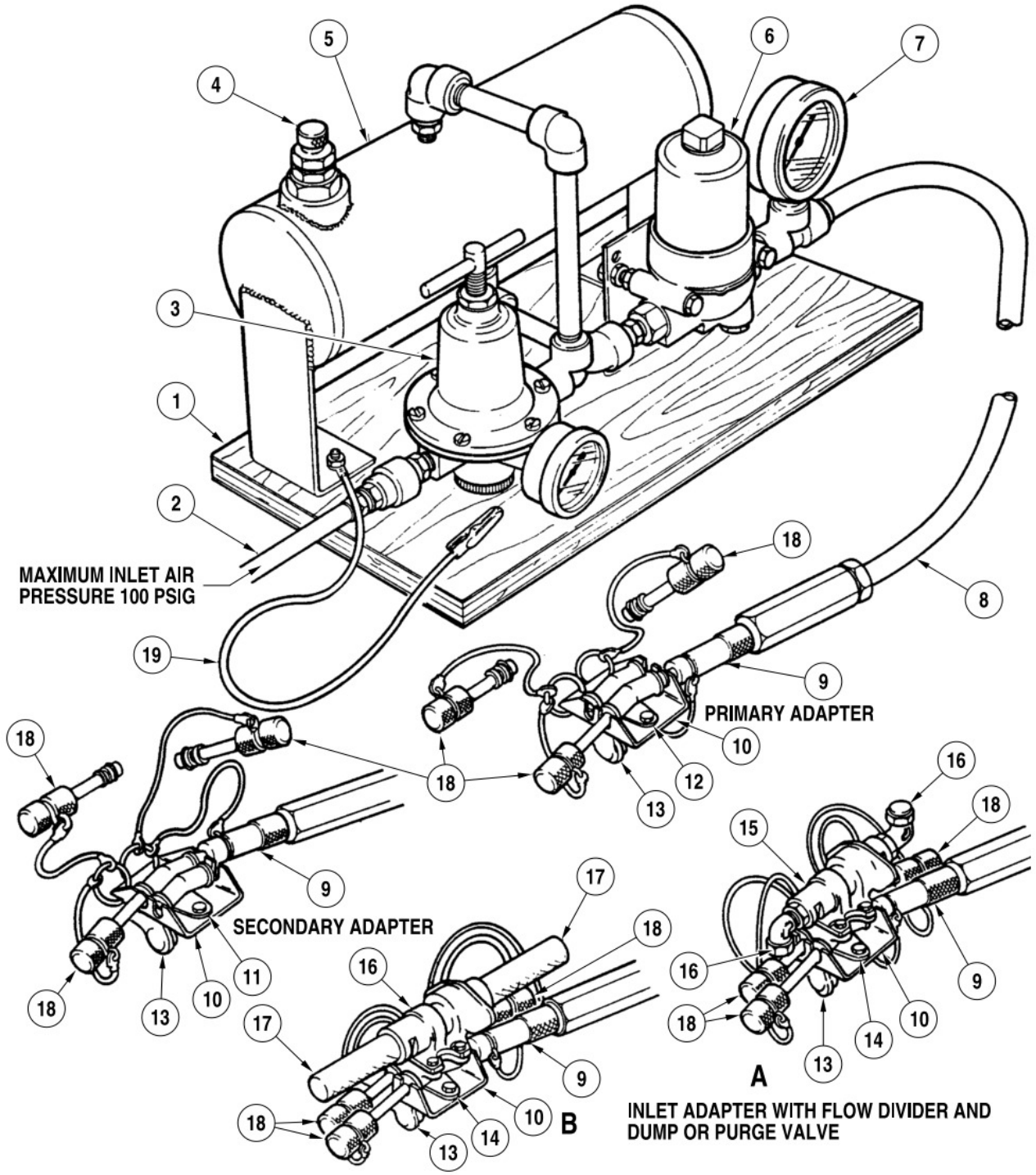
(7) Reduce pressure to fixture (2) to zero, disconnect hose assembly (1), and remove adapter assembly from fixture.

NOTE: If one or more unacceptable nozzle (s) is/are found, it is highly recommended that the hot section be inspected for damage by a direct visual inspection or borescope inspection.

C. Functional Test of Fuel Manifold Adapters with Test Rig (PWC30506) (Ref. Figs. 209 and 211)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 211 Manifold Adapter and Nozzle Assembly Functional Test



c1882i

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Test Rig (PWC30506)
2. Air Supply Line
3. Pressure Regulator (0 to 250 psig)
4. Relief Valve (150 psig)
5. Reservoir
6. Filter (10 micron nominal)
7. Pressure Gage (0 to 100 psig)
8. Hose Assembly
9. Connecting Tube Assembly
10. Nozzle Flow Fixture (PWC32811)
11. Primary Manifold Adapter
12. Secondary Manifold Adapter
13. Nozzle Assembly
14. Inlet Manifold Adapter
15. Flow Divider
16. Blanking Cap
17. Plug
18. Blanking Tube Assembly
19. Electrostatic Ground Cable

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: USE EXTREME CARE WHEN HANDLING FUEL NOZZLE ASSEMBLIES SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT FREE GLOVES SHOULD BE WORN AT ALL TIMES WHEN HANDLING PARTS.

NOTE: Although cleaning is recommended, nozzles can be reinstalled without cleaning provided that function test is within limits.

(1) Fill the reservoir (5) of the test rig (PWC30506) (1) with clean fuel (PWC01-001) or calibrating fluid (PWC03-002). Make sure that you do not fully fill the reservoir (5).

CAUTION: CONNECT GROUND CABLE ON TEST RIG TO ELIMINATE DANGER OF ELECTROSTATIC DISCHARGE.

CAUTION: AVOID BREATHING FUEL VAPOUR DURING FUNCTIONAL TEST. MAKE SURE OF ADEQUATE VENTILATION DURING TEST OR USE AN EXPLOSION PROOF VENTILATION HOOD.

(2) Functionally test the primary and secondary nozzle assemblies (13):

(a) Insert one blanking tube assembly (18) and connecting tube assembly (9) of fixture (10) into ports of the manifold adapter (11 and 12) as shown. The two remaining blanking tube assemblies (18) are not required.

NOTE: Primary adapters are identified by a black stripe, weld spot or the letter P on the adapter.

(b) Place manifold adapters (11 and 12) in fixture (10). Make sure the collars of tube assemblies (18 and 9) are located on the inside of fixture flanges with the tubes in the appropriate slots. Fasten adapter to fixture with screws and nuts.

(c) Connect hose assembly (8) to the tube assembly (9) and tighten coupling nuts securely to prevent leaks.

(d) Connect air supply line (2) with 100 psig maximum to pressure regulator (3) on test rig (1).

(e) Slowly adjust pressure regulator (3) to 20 psig. With the nozzle pointing downward, observe the spray pattern at the nozzle. An open spray must be observed, free from spitting or drooling.

(f) Adjust pressure regulator (3) to increase pressure to 60 psig as indicated on gage (7). Volume of spray should increase and be spread evenly about the center axis of nozzle orifice. If streakiness of more than 20% is evident, reject nozzle (Ref. Fig. 209).

NOTE: Spitting, drooling and streakiness may be caused by external carbon deposits around nozzle orifice. Remove deposits by lightly brushing nozzle face with cotton swab or non-metallic bristle brush while fuel is flowing through orifice.

(g) Return nozzles to an authorized accessories overhaul shop if satisfactory spray cannot be achieved after repeated cleaning.

(h) Reduce pressure to zero, as indicated on gage (7), by adjusting pressure regulator (3).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

When fuel flow from nozzle stops, disconnect and blank off hose assembly (8) from connecting tube assembly (10).

(i) Disconnect air supply line (2) from pressure regulator (3) on test rig (1).

(j) Remove manifold adapters (11 and 12) and nozzle assemblies (13) from flow fixture (11). Withdraw connecting tube assembly (10) and blanking tube assembly (19) from ports in adapter.

(3) Place manifold adapters (11 and 12) and nozzle assemblies (13) in a clean, dust-proof container until ready for installation on engine.

(4) Function test inlet manifold adapter (14) and nozzle assembly (13) :

(a) Blank off elbow connection on flow divider (15) with caps (16). Tighten caps 40 to 45 lb. in. (Ref. View A). Alternatively, if elbow connections are not installed, use plugs (17) to blank off ports (Ref. View B).

(b) Insert connecting tube assembly (9) of fixture (10) into port of inlet adapter (14) at same location as that used on secondary adapter (12).

(c) Insert three blanking tube assemblies (19) into remaining ports of adapter (14).

(d) Install adapter (14) in fixture (10) and test following Steps (2)(a) through (j).

(e) Remove plugs (17) or blanking caps (16) from flow divider (15). Place inlet manifold adapter and nozzle assembly with attached flow divider in a clean, dust proof container until required for installation.

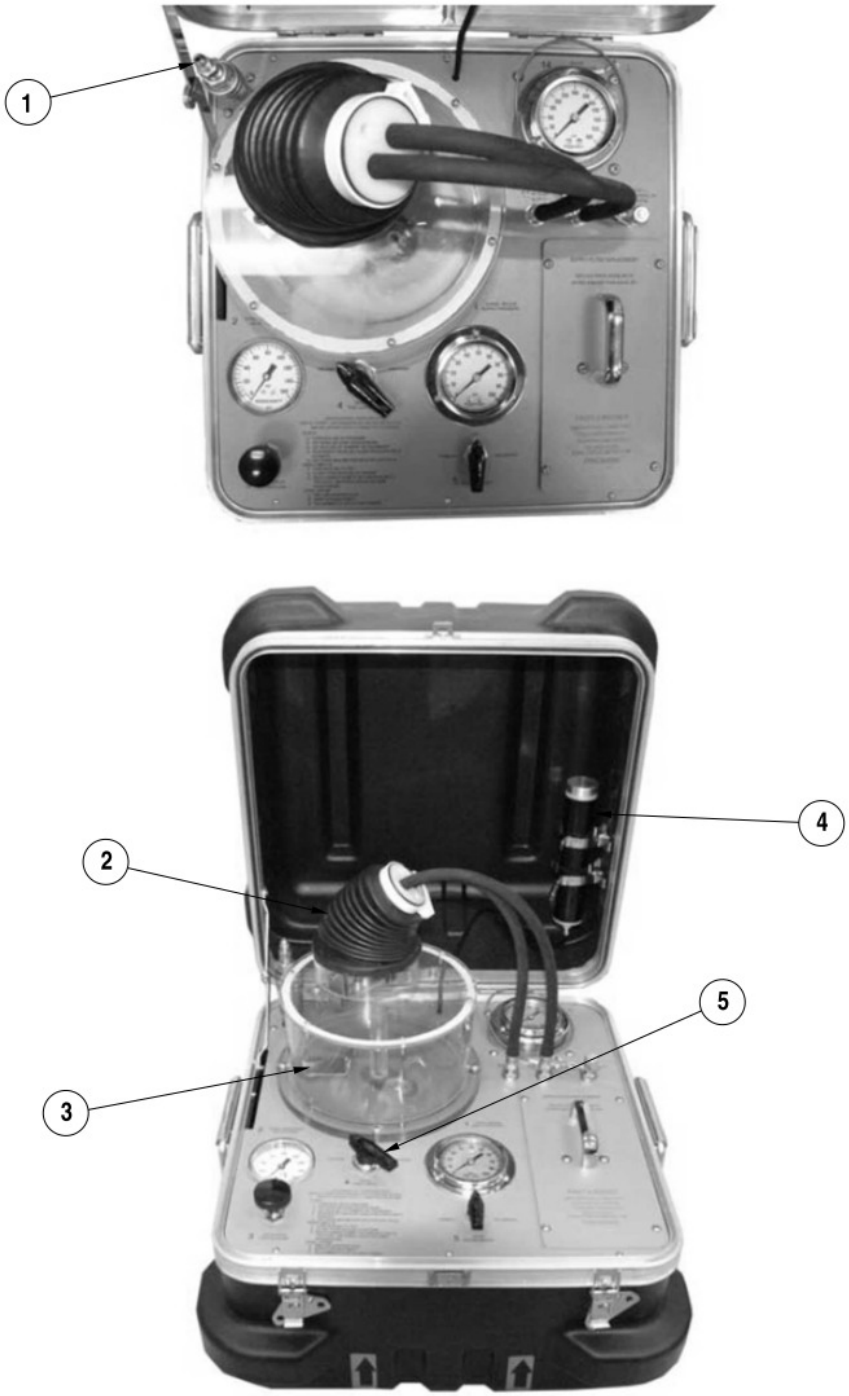
(5) If one or more nozzles are found unacceptable during the test, examine the hot section for evidence of heat damage by direct visual or borescope inspection.

NOTE: Return unserviceable nozzles to an authorized accessories overhaul shop for repair.

D. Functional Test of Fuel Manifold Adapters with Spray Chamber (PWC90086) (Ref. Fig. 212)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 212 Spray Chamber (PWC90086)



c209883

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Air Inlet
2. Rubber Boot
3. Spray Chamber
4. Battery Pack
5. Selector Valve

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: TAKE EXTREME CARE WHEN HANDLING FUEL NOZZLE ASSEMBLIES SINCE EVEN FINGERPRINTS ON THE ORIFICE MAY PRODUCE POOR SPRAY PATTERN. CLEAN, LINT FREE GLOVES MUST BE USED AT ALL TIMES WHEN HANDLING PARTS.

NOTE: Although cleaning is recommended, nozzles can be reinstalled without cleaning provided that function test is within limits.

(1) Fill the reservoir of the spray chamber (PWC90086) with clean fuel (PWC01-001) or calibrating fluid (PWC03-002). Make sure that you do not fully fill the reservoir.

CAUTION: CONNECT GROUND CABLE ON SPRAY CHAMBER (PWC90086) TO PREVENT DANGER OF ELECTROSTATIC DISCHARGE.

(2) Attach ground strap clip to a suitable ground.

(3) Connect air supply line with 100 psig maximum to air inlet (1).

(4) Put fuel nozzle assembly on positioning fixture.

(5) Connect nozzle inlet connector to fuel nozzle assembly.

(6) Position locking mechanism over fuel nozzle inlet connection to make sure that the fuel nozzle is positioned over light source.

(7) Install rubber boot (2) over the fuel nozzle and spry chamber (3) to reduce any possible calibration fluid vapor escaping from the chamber.

(8) Turn on light source at the switch on the battery pack (4) mounted on the top lid.

(9) Set initial air pressure to 20 psig.

(10) Set selector valve (5) in "Pressure" position.

(11) Inspect the fuel spray cone at 20 psig for drips and voids.

(12) Adjust pressure regulator valve to increase pressure to 60 psig and inspect fuel nozzle spray cone for drips and voids.

NOTE: Spitting, drooling and streakiness may be caused by external carbon deposits around nozzle orifice. Remove deposits by lightly brushing nozzle face with cotton swab or non-metallic bristle brush while fuel is flowing through orifice.

(13) Return fuel nozzles to an authorized accessories overhaul shop if satisfactory spray cannot be achieved after repeated cleaning.

(14) After completion of testing, move selector valve (5) to "Vacuum" position to remove the accumulated calibration fuel from the spray chamber (3).

NOTE: The calibration fuel is filtered and returned to the reservoir.

(15) Reduce pressure to zero, as indicated on gage by adjusting pressure regulating valve.

(16) Disconnect air supply line from spray chamber (PWC90086).

(17) Remove fuel nozzle from spray chamber (PWC90086).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(18) Put nozzle assemblies in a clean, dust proof container until ready for installation on engine.

(19) If one or more nozzles are found unacceptable during the test, examine the hot section for evidence of heat damage by direct visual or borescope inspection.

NOTE: Return unserviceable nozzles to an authorized accessories overhaul shop for repair.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement